

Course Overview

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Healthcare Data Privacy and Security

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S.K.I.L.L.E.D — AID013244/04/2

Strategic Knowledge and Inclusive Lifelong Learning in the health sector for youth Employability and Development

Roadmap I



1 Course Overview

2 Program

About the Course I

Healthcare Data Privacy and Security introduces students and healthcare professionals to the principles, policies, and technologies used to protect health data in digital systems.

Topics covered:

- Security foundations, risk, and healthcare threats
- Cryptography, authentication, access control, and networking
- Privacy regulation, operational security, and incident response
- Applied labs using notebooks and OpenHospital

Schedule: 9 lessons × 3h30m each

Running Scenario

St. Isidore Hospital

The course repeatedly returns to the same fictional hospital environment: clinical workstations, laboratory systems, patient records, OpenHospital, remote access, guest networks, backups, logs, and privacy obligations.

Why this helps

Each lesson adds one layer to the same hospital story, so technical controls and privacy duties do not feel like disconnected lists.

Lesson 1 — Security Foundations

- Why healthcare security matters
- Confidentiality, integrity, and availability
- Risk, likelihood, impact, and basic risk matrices
- Threats, vulnerabilities, attacks, and attack surfaces
- Running example: St. Isidore Hospital

Outcome: describe security goals and risks for a healthcare system.

Lesson 2 — Cryptographic Foundations

- Classical encryption as intuition
- Symmetric encryption: DES, 3DES, AES
- Block ciphers, stream ciphers, and modes
- Authenticated encryption, hashes, HMAC, password hashing
- RSA, signatures, certificates, key exchange, randomness
- Python notebooks with concrete examples

Outcome: understand what cryptographic tools do and where they fit.

Lessons 3–4 — Access Control and OpenHospital



- Subjects, objects, rights, and policies
- DAC, MAC, RBAC, least privilege, separation of duties
- Python notebooks comparing access-control models
- Deploy and access OpenHospital
- Configure users, groups, permissions, and role behavior
- Inspect logs and identify audit gaps

Outcome: design, implement, and verify healthcare access-control behavior.

Lesson 5 — Authentication

- Authentication factors
- Password files, password storage, attacks, and policies
- Challenge-response authentication
- Tokens, smart cards, remote authentication, and MFA
- One-time passwords and TOTP
- Python notebooks on password hashing, challenge-response, and MFA

Outcome: understand common authentication mechanisms and their failure modes.

Lessons 6–7 — Networking and API Security

- IP, subnets, ports, DNS, routing, and hospital network architecture
- Firewalls, packet filtering, stateful firewalls, proxies, and VPNs
- Hands-on firewall policy simulation
- Deploy the real OpenHospital API with MariaDB
- Put Nginx in front of the API
- Harden access by subnet and API path, then verify with logs

Outcome: reason about segmentation and test a realistic healthcare API proxy.

Lesson 8 — Privacy and Data Protection

- From privacy as private life to data protection as information governance
- Personal data, medical data, PHI, identifiability
- Anonymization and pseudonymization
- GDPR as an influential reference model, not the only global regime
- Consent, legal bases, patient rights, accountability, DPIA, breach duties
- Links between privacy duties and technical safeguards [Conduah et al., 2025]

Outcome: connect technical safeguards with legal and ethical duties.



- Operational security mindset
- Phishing, scams, fake IT support, invoice fraud, MFA fatigue
- Ransomware attack chain and healthcare impact
- Backups, 3-2-1 strategy, immutable/offline copies
- RPO, RTO, restore testing, and downtime procedures
- Incident response, breach triage, communication, and tabletop exercise

Outcome: describe practical actions before, during, and after an incident.

Course Integration

- ① Security goals and risk
- ② Encryption and integrity mechanisms
- ③ Access-control models
- ④ Real OpenHospital RBAC configuration
- ⑤ Authentication mechanisms and attacks
- ⑥ Networking and firewall policy
- ⑦ Real OpenHospital API proxy lab
- ⑧ Privacy and data protection governance
- ⑨ Ransomware, backups, incident response, and communication

Labs and Practical Material

- Cryptography notebooks
- DAC/MAC/RBAC notebooks
- Authentication notebooks
- Firewall simulation notebook
- OpenHospital RBAC lab
- OpenHospital API network/proxy lab
- Operational tabletop exercise

Optional Extensions



- Health information systems and interoperability: EHR, LIS, RIS, PACS, DHIS2, HL7/FHIR
- Standards and frameworks: ISO 27001, NIST, CIS Controls
- AI and health data: privacy-preserving analytics, inference risks, synthetic data
- Medical devices, wearables, mobile health, and telemedicine security

Doubts? Mail me! I



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References I



- [Conduah et al., 2025] Conduah, A., Ofoe, S., and Siaw-Marfo, D. (2025).
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